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SKG 12 Mechanical Engineering

Predictive Maintenance Technique – Part 1 (Failure Mechanism), L2

Failure Mechanisms and Detection Parameters (4 of 4)

Electrical and Instrument Control Equipment

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Failure Mechanisms and Detection Parameters for Electrical and Instrument Control Equipment

Learning Objectives

By the end of this session, you should be able to:

- Understand the various degradation mechanisms affecting equipment components i.e. RCFA approach
- Understand the failure physics relating the degradation parameters for specific failure mechanism i.e. the deterministic approach, to be used in RCFA and detecting incipient failures through Predictive Maintenance Techniques.
- Aware of the re-engineering and RCFA approach that can be implemented to design out or avoid recurrence of failure.
- Aware to the various technology for detection of degradation parameters, and interpretation of results.

Failure Mechanisms related following Rotating Machinery Covered

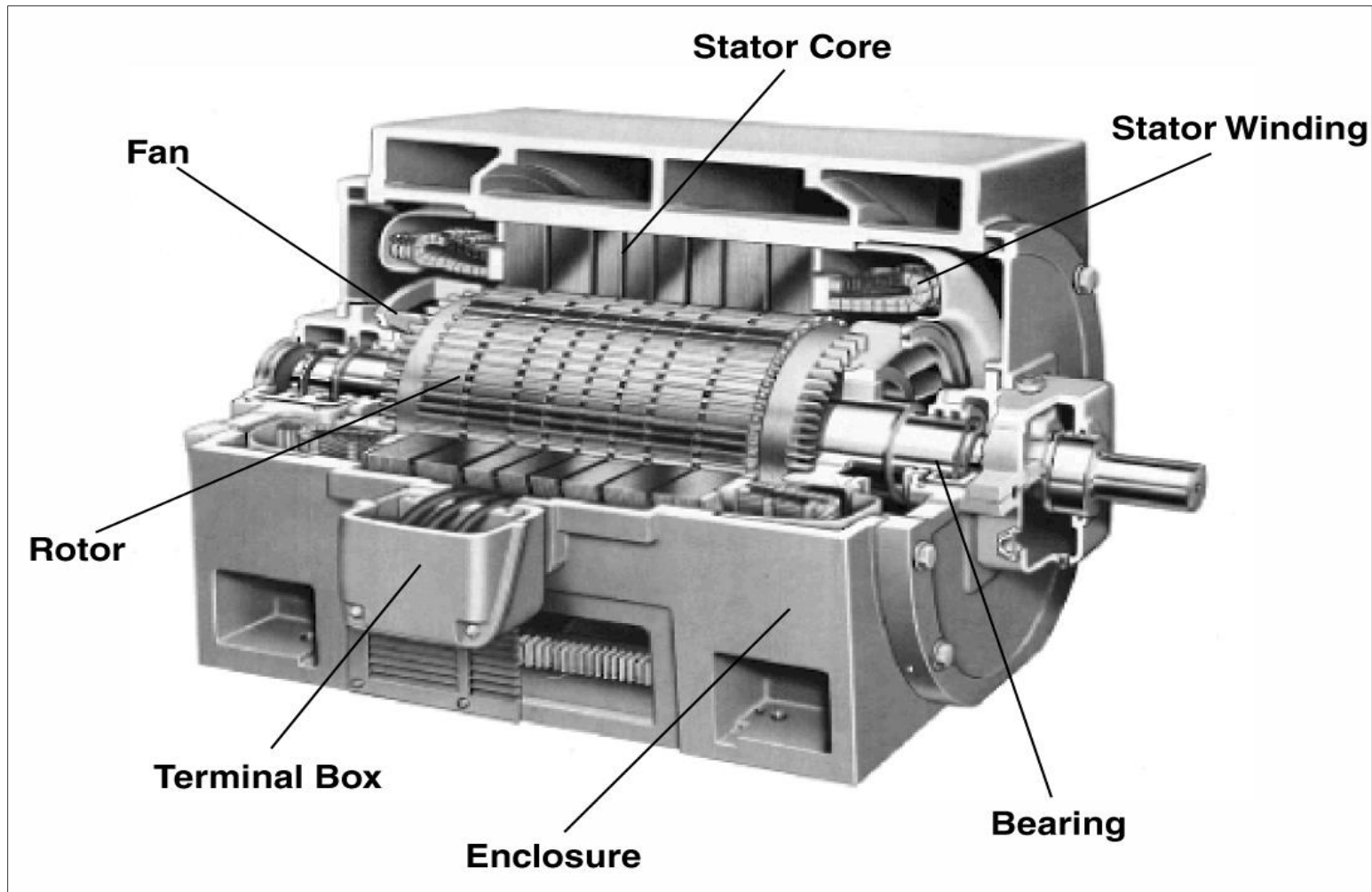
- **Electric Motor**
- **Power Transformer**
- **Uninterrupted Power Supply**
- **Valves**

Pre-requisite Information

Technical drawings/diagrams containing process data, material and media codes:

- Single line Diagram
- PFD & P&IDs, Equipment Data Sheet
- Shut down logic if any
- Cause and effect diagrams if applicable
- Fire protection data sheets if applicable
- Asset Structure / Acceptance Criteria

Electric Motor



Electric Motor

Sno	Component	Failure Mechanism	Failure Mode	Detection Parameter
01	Rotor/Stator	Solid dielectric insulation degradation	Short circuit. Motor trips off line.	1. Dielectric properties (Insulation resistance and polarization index). 2. Winding Temperature 3. Motor Current Analysis
02	Bearing(s)	Wear out	Noise, excessive vibration or heat. Loss of shaft support, rubbing or seizure of rotating components, possible shaft fracture.	1. Vibration measurement 2. Noise measurement 3. Frequency measurement thru Ultrasonic Ins
03	Bearing(s)	Lubrication system failure	Noise, excessive vibration or heat. Loss of shaft support, rubbing or seizure of rotating components, possible shaft fracture.	1. Bearing Temperature measurement. 2. Noise and Vibration measurement. 3. LO Properties / Wear debris Analysis

Electric Motor

SNo	Component	Failure Mechanism	Failure Mode	Detection Parameter
04	Terminal Box	Solid dielectric insulation degradation	Short circuit. Motor trips off line.	1. Insulation Resistance. 2. Earth Leakage / Grounding 3. Heat Release measure thru IR detector
05	Gasket	Gasket degradation	Leak allows water ingress resulting in short circuit. Motor trips off line.	1. Insulation Resistance 2. Earth Leakage / Grounding
06	Instrumentation	Calibration drift	Trips motor off line.	1. Electrical Performance Parameter Monitoring. 2. Vibration / Noise Measurement
07	Instrumentation	Contact corrosion	Protective device fails to operate.	1. Earth Leakage / Ground Resistance / Impedance. 2. Electrical Performance Parameter Monitoring.

Electric Motor

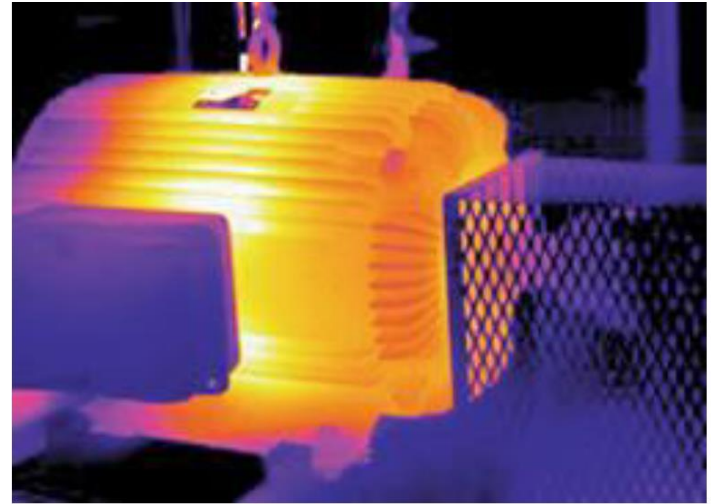
Failure Mechanism

1. Power quality
2. Power circuit
3. Insulation
4. Stator
5. Rotor
6. Air gap

Six Failures sources in Motor



Typical Bearing Failure in Motor

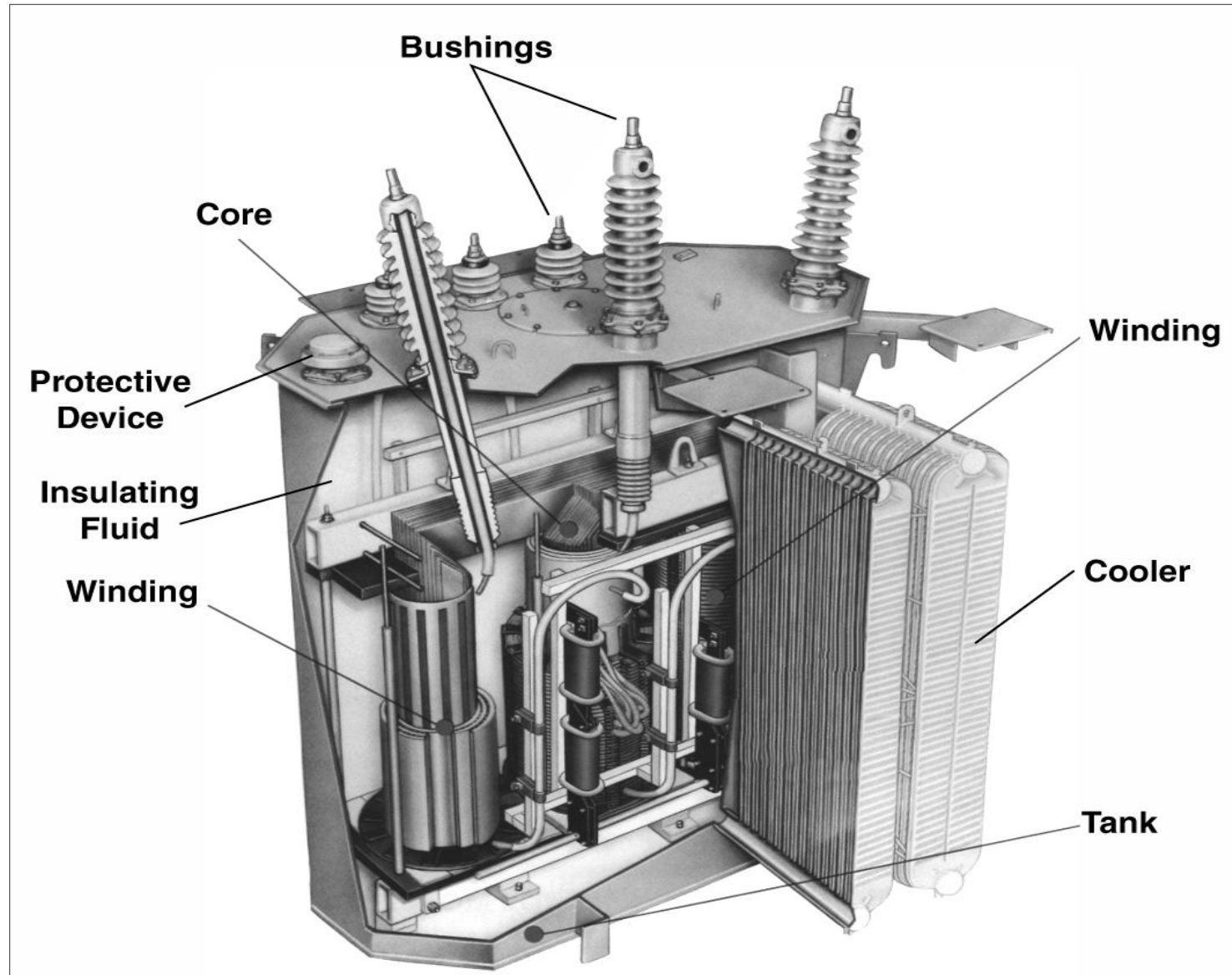


Typical Motor Winding Failure detected by Heat Energy using IR



Prolonged Bearing defect lead to Rotor Failure

Power Transformer

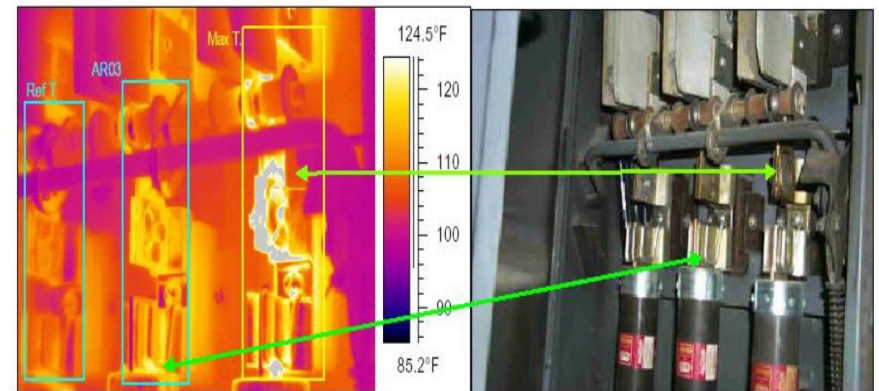
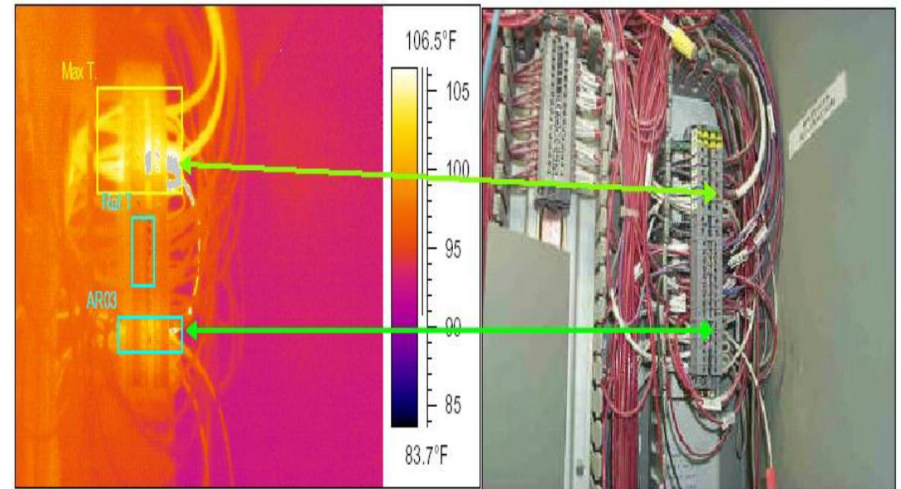
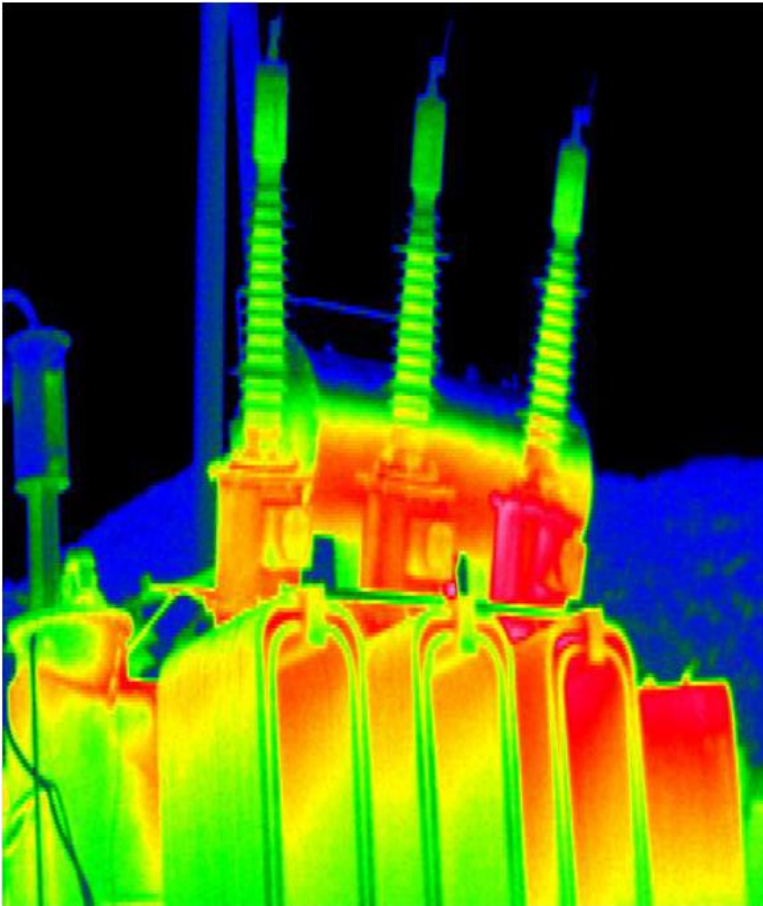


Power Transformer

S No	Component	Degradation Mechanism	Failure Mode	Detection Parameter
01	Windings	Solid dielectric insulation degradation	Short circuit. Transformer trips off line.	1. Resistance between Phase-Phase / Phase - Ground. 2. Electrical Performance Parameter Monitoring
02	Tank/Coolers	Corrosion/Rust	Leak. Transformer trips off line.	1. Loss of Thickness 2. Leak rate
03	Insulating fluid	Insulating fluid degradation	Deterioration of insulating properties results in a short circuit. Transformer trips off line.	1. Oil Properties including Moisture Content, O2 Presence, Sludge. 2. Heat Energy thru IR
04	Instrumentation	Calibration drift	Transformer trips off line.	1. Electrical Parameter Monitoring 2. Noise Measurement
05	Instrumentation	Contact corrosion	Protective device fails to operate.	1. Earth Leakage / Ground Resistance / Impedance. 2. Electrical Performance Parameter Monitoring

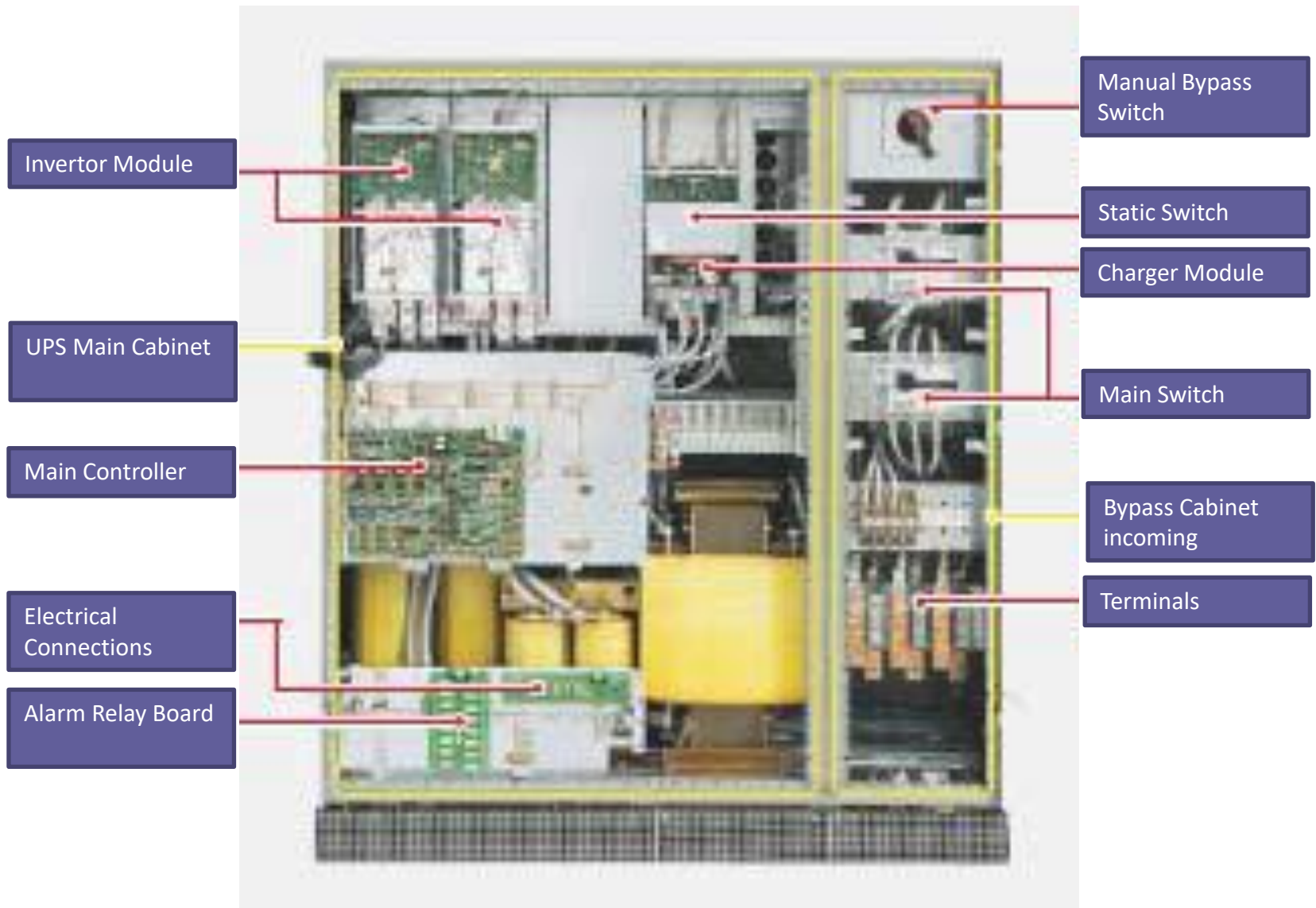
Power Transformer

Failure Mechanism

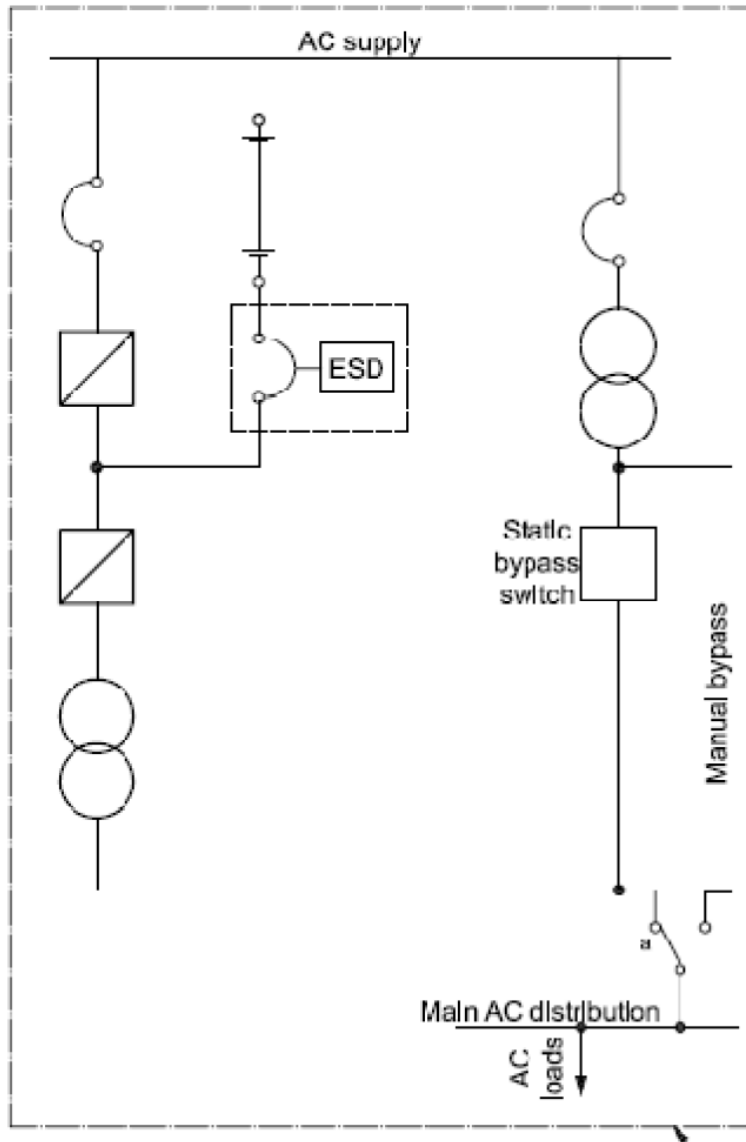


Above figures show Transformed under Load Condition and Status of Various Cable glands, Instrumentation Cable utilizing Infra Red Thermography (Detection Parameter Heat Dissipation Measurement)

UPS



UPS



Applicable items:

- UPS systems and batteries
- Components: rectifier, inverter, static switch, capacitor, fan, cards.

UPS Functional Requirements:

AC UPS: To generate uninterruptible electrical power supply in the range of 64 kW at rated voltage of 240V AC and at frequency of 50 Hz.

DC UPS: To generate uninterruptible electrical power supply in the range of ampere at rated voltage of 24V/110V DC.

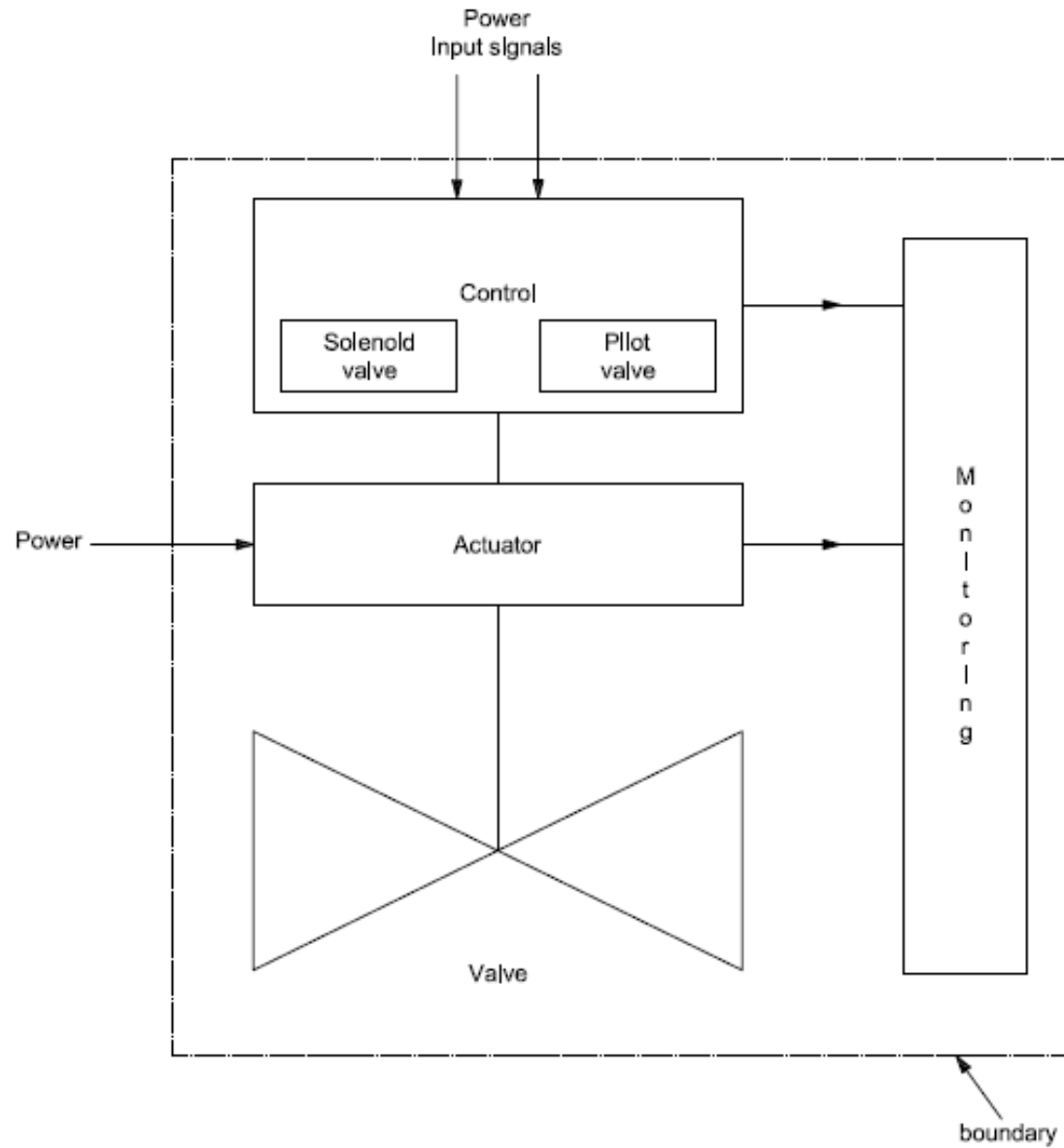
UPS

Component	Failure Mode	Failure Mechanism	Description	Detection Parameter
Batteries	Fault alarm indication such as earth fault, inverter/rectifier fault	Battery terminal dirty or contaminated	Battery terminal dirty\contaminated due to lack of maintenance on cleanliness of battery terminal and other terminations.	Battery Condition
Batteries	UPS battery fail to back up	Battery failure	Battery failure (i.e. leak, bulging, low electrolyte, dry, terminal corrosion, H ₂ S smell).	Battery Condition
Cables	Burning smell	Cable or wiring melted	Cable/wiring melted due to material defect	Cable condition
Cables terminal	Fault alarm indication such as earth fault, inverter/rectifier fault	Cable loose	Cable loose connection inside the compartment due to error during servicing/maintenance.	Resistance across cable
Capacitors	No output voltage at inverter (240V)	Electronic component fail	Electronic component fail (capacitor blown/leak or bulging) due to material defect (prolong used) and static charge during handling.	Capacitance

UPS

Component	Failure Mode	Failure Mechanism	Description	Detection Parameter
Capacitors	Fault alarm indication such as earth fault, inverter/rectifier fault	Capacitor fluid leaking or bulging	Capacitor fluid leaking or bulging due to negligence in replacing capacitor after it reaches operating life time.	Capacitance / Capacitor condition
Cards backplane	No output voltage at inverter (240V)	Control card loosen	Control card loosen from the slot holder due to vibration from platform or movement of nearby equipment i.e. from Gas Compressor, Gas Turbine, etc.	Capacitance / Backplane condition
Communication Cards	Indication LCD total blackout	Communication card shorted	Communication card shorted due to dust accumulation due lack of maintenance on cleanliness.	Communication Card Condition
Communication Cards	Indication LCD total blackout	Communication card dry joint problem	Communication card dry joint problem due to material defect	Communication Card Condition
Electronic Cards	No output voltage at	Inverter card shorted	Inverter card shorted due to dust	Capacitance / Backplane condition

Valves



Valves

Component	Failure Mode	Failure Mechanism	Description	Detection Parameter
stem, packing, gaskets	Valve components leak process fluid	Wear out - valve components	Leaking of process liquid	Visually inspect for leaks
Valve stem	erodes in two	Induced - Flashing or cavitation	Cavitation of the valve (applicable for steam service only)	Listen for valve cavitation and vibration
Valve actuator seals	Valve components leak process fluid	Wear out - Gasket fatigue	Detection of leaking of process liquid	Inspect valve and listen for external air leaks
Actuator	Valve passing	Wear out - fatigue	Valves that leak internally	Acoustic meter, Infrared



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Thank you